

Chemical Bonding and Molecular Structure

Covalent Bond, Electrovalent Bond, Lattice Enthalpy and Octet Rule

- Which of the following species contains equal number of σ and π -bonds? (2015 Re)
 - (CN)₂
 - (CH)₂(CN)₂
 - HCO₃⁻
 - XeO₄
- Which one of the following molecules contains no π bond? (2013)
 - SO₂
 - NO₂
 - CO₂
 - H₂O
- Which of the following is electron-deficient? (2013)
 - (CH₃)₂
 - (SiH₃)₂
 - (BH₃)₂
 - PH₃
- The H—C—H bond angle in CH₄ is larger than the H—N—H bond angle in NH₃
- The H—C—H bond angle in CH₄, the H—N—H bond angle in NH₃, and the H—O—H bond angle in H₂O are all greater than 90°.
- Then H—O—H bond angle in H₂O is larger than the H—C—H bond angle in CH₄
- The H—O—H bond angle in H₂O is smaller than the H—N—H bond angle in NH₃
- Which of the following molecules has the maximum dipole moment? (2014)
 - CH₄
 - NH₃
 - NF₃
 - CO₂

VSEPR Theory

Bond Parameters and Dipole Moment

- Amongst the following, which one will have maximum 'lone pair-lone pair' electron repulsions? (2022)
 - XeF₂
 - ClF₃
 - IF₅
 - SF₄
 - Match List-I with List-II.

List-I		List-II	
(A)	PCl ₅	(i)	Square pyramidal
(B)	SF ₆	(ii)	Trigonal planar
(C)	BrF ₅	(iii)	Octahedral
(D)	BF ₃	(iv)	Trigonal bipyramidal

Choose the correct answer from the options given below. (2021)

 - A-ii B-iii C-iv D-i
 - A-iii B-i C-iv D-ii
 - A-iv B-iii C-ii D-i
 - A-iv B-iii C-i D-ii
 - Identify the wrongly matched pair. (2020-Covid)

Molecule	Shape or geometry of molecule
a. SF ₆	Octahedral
b. BeCl ₂	Linear
c. NH ₃	Trigonal pyramidal
d. PCl ₅	Trigonal planar
- Which of the following molecules is non-polar in nature? (2021)
 - CH₂O
 - SbCl₅
 - NO₂
 - POCl₃
 - Which of the following set of molecules will have zero dipole moment? (2020)
 - Boron trifluoride, hydrogen fluoride, carbon dioxide, 1,3-dichlorobenzene
 - Nitrogen trifluoride, beryllium difluoride, water, 1,3-dichlorobenzene
 - Boron trifluoride, beryllium difluoride, carbon dioxide, 1,4-dichlorobenzene
 - Ammonia, beryllium difluoride, water, 1,4-dichlorobenzene
 - Which one of the following pair of species have the same bond order? (2017-Delhi)
 - N₂, O₂⁻
 - CO, NO
 - O₂, NO⁺
 - CN⁻, CO
 - Consider the molecules CH₄, NH₃ and H₂O. Which of the given statement is false? (2016 - I)



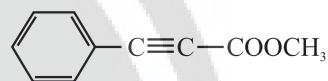
12. Identify the incorrect statement related to PCl_5 from the following: (2019)
- Three equatorial P-Cl bonds make an angle of 120° with each other
 - Two axial P-Cl bonds make an angle of 180° with each other
 - Axial P-Cl bonds are longer than equatorial P-Cl bonds
 - PCl_5 molecule is non-reactive
13. The species, having bond angles of 120° is (2017-Delhi)
- BCl_3
 - PH_3
 - ClF_3
 - NCl_3
14. Among the following which one is a wrong statement? (2016 - II)
- SeF_4 and CH_4 have same shape
 - I_3^+ has bent geometry
 - PH_5 and BiCl_5 do not exist
 - $p\pi - d\pi$ bonds are present in SO_2
15. Which of the following pairs of ions is isoelectronic and isostructural? (2016-II)
- CO_3^{2-} , NO_3^-
 - ClO_3^- , CO_3^{2-}
 - SO_3^{2-} , NO_3^-
 - ClO_3^- , SO_3^{2-}
16. Predict the correct order among the following: (2016 - I)
- Lone pair — bond pair > bond pair — bond pair > lone pair — lone pair
 - Lone pair — lone pair > lone pair — bond pair > bond pair — bond pair
 - Lone pair — lone pair > bond pair — bond pair > lone pair — bond pair
 - Bond pair — bond pair > lone pair — bond pair > lone pair — lone pair
17. Maximum bond angle at nitrogen is present in which of the following? (2015)
- NO_2^+
 - NO_3^-
 - NO_2
 - NO_2^-
18. In which of the following pairs, both the species are not isostructural? (2015)
- Diamond, Silicon carbide
 - NH_3 , PH_3
 - XeF_4 , XeO_4
 - SiCl_4 , PCl_4^+
19. Which of the following pairs of ions are isoelectronic and isostructural? (2015)
- SO_3^{2-} , NO_3^-
 - ClO_3^- , SO_3^{2-}
 - CO_3^{2-} , SO_3^{2-}
 - ClO_3^- , CO_3^{2-}
20. Which one of the following species has planar triangular shape? (2014)
- NO_3^-
 - NO_2^-
 - CO_2
 - N_3^-
21. Which of the following is a polar molecule? (2013)
- BF_3
 - SF_4
 - SiF_4
 - XeF_4
22. XeF_2 is isostructural with (2013)
- SbCl_3
 - BaCl_2
 - TeF_2
 - ICl_2^-

Valence Bond Theory, Hybridisation

23. BF_3 is planar and electron deficient compound. Hybridization and number of electrons around the central atom, respectively are: (2021)

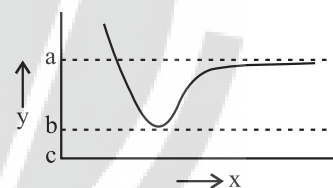
- sp^3 and 6
- sp^2 and 6
- sp^2 and 8
- sp^3 and 4

24. How many (i) sp^2 hybridised carbon atoms and (ii) π bonds are present in the following compound? (2020-Covid)



- 8, 6
- 7, 6
- 8, 5
- 7, 5

25. The potential energy (y) curve for H_2 formation as a function of internuclear distance (x) of the H atoms is shown below. (2020-Covid)



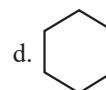
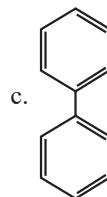
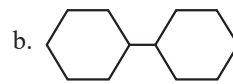
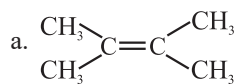
The bond energy of H_2 is

- $\frac{(c-a)}{2}$
- $\frac{(b-a)}{2}$
- $(c-a)$
- $(b-a)$

26. The hybridisations of atomic orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ respectively are: (2016-II)

- sp , sp^3 and sp^2
- sp^2 , sp^3 and sp
- sp , sp^2 and sp^3
- sp^2 , sp and sp^3

27. In which of the following molecules, all atoms are coplanar? (2016 - II)



33. Consider the following species: (2018)
 CN^+ , CN^- , NO and CN
 Which one of these will have the highest bond order?
 a. NO b. CN^-
 c. CN d. CN^+
34. Decreasing order of stability of O_2 , O_2^- , O_2^+ and O_2^{2-} is: (2015 Re)
 a. $\text{O}_2^- > \text{O}_2^{2-} > \text{O}_2^+ > \text{O}_2$
 b. $\text{O}_2^+ > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$
 c. $\text{O}_2^{2-} > \text{O}_2^- > \text{O}_2 > \text{O}_2^+$
 d. $\text{O}_2 > \text{O}_2^+ > \text{O}_2^{2-} > \text{O}_2^-$
35. Which of the following options represents the correct bond order? (2015)
 a. $\text{O}_2^- < \text{O}_2 < \text{O}_2^+$ b. $\text{O}_2^- > \text{O}_2 < \text{O}_2^+$
 c. $\text{O}_2^- < \text{O}_2 > \text{O}_2^+$ d. $\text{O}_2^- > \text{O}_2 > \text{O}_2^+$
36. The correct bond order in the following species is: (2015)
 a. $\text{O}_2^{2+} < \text{O}_2^- < \text{O}_2^+$ b. $\text{O}_2^{2+} < \text{O}_2^- < \text{O}_2$
 c. $\text{O}_2^- < \text{O}_2^+ < \text{O}_2^{2+}$ d. $\text{O}_2^{2+} < \text{O}_2^+ < \text{O}_2^-$
37. Which of the following is paramagnetic? (2013)
 a. CO b. O_2^-
 c. CN^- d. NO^+

Hydrogen Bonds

- 30.** Which amongst the following is incorrect statement (2022)
- O_2^+ ion is diamagnetic.
 - The bond orders of O_2^+ , O_2 , O_2^- and O_2^{2-} are 2.5, 2, 1.5 and 1, respectively
 - C_2 molecule has four electrons in its two degenerate π molecular orbitals.
 - H_2^+ ion has one electron.
- 31.** Identify a molecule which does not exist. (2020)
- Li_2
 - C_2
 - O_2
 - He_2
- 32.** Which of the following diatomic molecular species has only π bonds according to Molecular Orbital Theory? (2019)
- O_2
 - N_2
 - C_2
 - Be_2
- 33.** The correct bond order in the following species is: (2015)
- $O_2^{2+} < O_2^- < O_2^+$
 - $O_2^{2+} < O_2^- < O_2^+$
 - $O_2^- < O_2^+ < O_2^{2+}$
 - $O_2^{2+} < O_2^+ < O_2^-$
- 34.** Which of the following is paramagnetic? (2013)
- CO
 - O_2^-
 - CN $^-$
 - NO $^+$
- 35.** Which one of the following compounds shows the presence of intramolecular hydrogen bond? (2016 - II)
- Cellulose
 - Concentrated acetic acid
 - H_2O_2
 - HCN
- 36.** The correct bond order in the following species is: (2015)
- $O_2^{2+} < O_2^- < O_2^+$
 - $O_2^{2+} < O_2^- < O_2^+$
 - $O_2^- < O_2^+ < O_2^{2+}$
 - $O_2^{2+} < O_2^+ < O_2^-$
- 37.** Which of the following is paramagnetic? (2013)
- CO
 - O_2^-
 - CN $^-$
 - NO $^+$
- 38.** Which one of the following compounds shows the presence of intramolecular hydrogen bond? (2016 - II)
- Cellulose
 - Concentrated acetic acid
 - H_2O_2
 - HCN

Answer Key

[illegible]